

2024 REPORT

State of Network Threat Detection



ARISTA

ENEA

 CUSTOCY®

Executive Summary

While “platformization” has been a hot topic in 2024, it has also been a year in which security professionals have looked to advanced, highly specialized tools to help them solve thorny problems that not only persist but seem to grow more challenging by the day. Among these are acute alert fatigue, a steady erosion of network visibility, and a growing sophistication in cyberattacks.

Among the specialized tools security professionals are looking to are Network-based Threat Detection (NTD) solutions, such as Network-based Intrusion Detection Systems (NIDS) and Network-based Threat Detection and Response (NDR). To better understand the state of Network Threat Detection and whether today’s solutions and supporting technologies— like deep packet inspection— are meeting contemporary security challenges, Cybersecurity Insiders surveyed its 600,000-member information security community. The survey reveals that while NTD tools are widely deployed and positively viewed, they must evolve if they are going to help security professionals meet significant present-day and emerging challenges.

Key findings:

ALERT ISSUES

- **Alert prioritization** is the #1 overall operational challenge for security teams
- **Alert accuracy & actionability** is cited as the greatest challenge with NIDS specifically

VISIBILITY CHALLENGES

- **No (or poor) global attack surface visibility** is the #2 overall operational challenge
- **Encrypted traffic** is the #1 network blind spot, which 55% report negatively impacts security

DESIRED PRODUCT ENHANCEMENTS

- **AI integration:** 71% consider AI integration extremely or very important for combatting advanced threats
- **Automatic scoring & prioritization of threats** named the #1 must-have for an effective network threat detection solution

DEPLOYMENT PLANS & PREFERENCES

- **Majority (66%) plan to implement anomaly detection** over the next 6 to 24 months; only 17% report having an NTD solution now that uses anomaly detection
- **Majority (59%) prefer standalone NTD solutions** (DPI sensor, NIDS, NDR, XDR) to NTD within multi-function security platforms (e.g., SASE, SSE)

Experts from Enea, Arista Security, and Custocy discuss options and strategies for addressing the needs and concerns raised in this survey in a panel discussion. We invite you to watch the webinar “2024 State of Network Threat Detection” on November 14, 2024, or afterwards on-demand.

Many thanks to [Enea](#), [Arista Security](#) and [Custocy](#) for supporting this important research project, with special gratitude to Enea for their invaluable contribution to this report.

Holger Schulze

Founder, Cybersecurity Insiders

Table of Contents

FINDINGS

Awareness	4
Challenges	5
Enhancements	13
Plans	16
Preferences	17

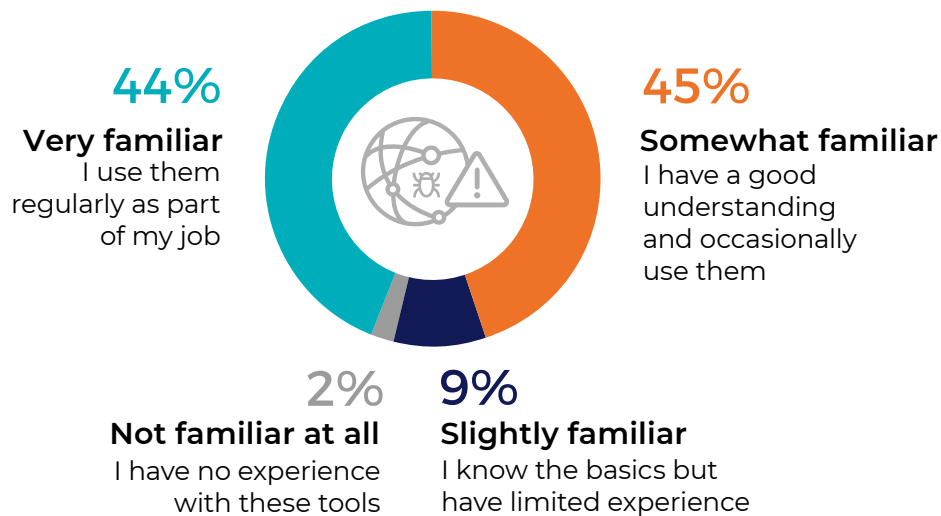
THE SURVEY

Methodology & Demographics	20
----------------------------	----

Even Split on Familiarity with & Opinion of NTD

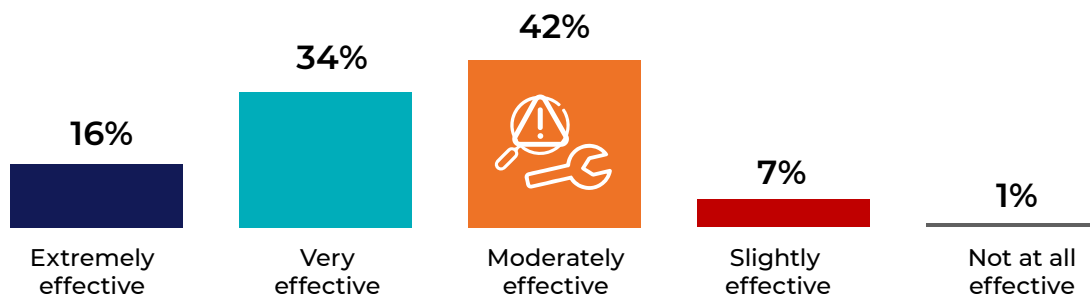
About half of respondents (44%) are very familiar with NTD tools and use them regularly, while a similar percentage (45%) are only somewhat familiar with them and use them only occasionally. The rest are only slightly familiar, or not familiar at all, with NTD tools.

► How familiar are you with network threat detection tools and technologies?



A similar breakdown applies to the perceived effectiveness of NTD solutions: half (50%) rate them as either extremely or very effective while 42% find them only moderately effective, and 8% find them slightly or not at all effective. While differences in domain specialization may affect awareness and usage, all security team members would benefit from increased awareness of the vital role NTD plays in contemporary multilayered defensive systems. With regard to confidence levels, much progress can be made by focusing solution roadmaps on the important challenges identified in this survey.

► How effective do you think network threat detection is in mitigating risk and reducing your risk profile?

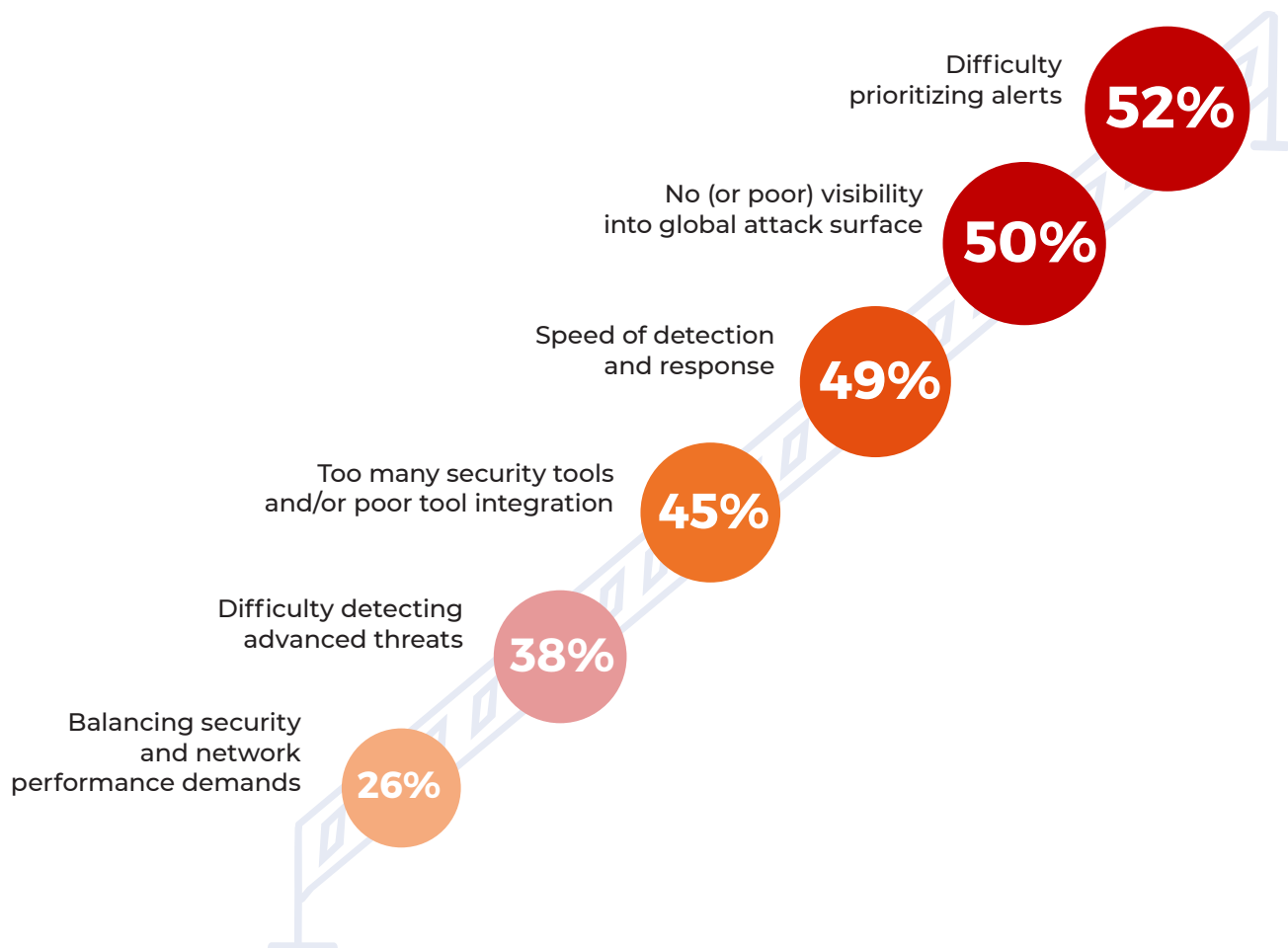


Alerts & Visibility Are Top Operational Challenges

When asked for their top three operational challenges, the difficulty prioritizing alerts emerged as a top challenge for 52% of respondents. Given the huge volume of alerts frontline security professionals typically face, distinguishing between critical and low-risk incidents can be a major (and highly frustrating) hurdle.

This issue is compounded by a lack of visibility into the global attack surface (50%), which opens a crucial gap in defensive capabilities as organizations expand into cloud and hybrid environments, the number of edge locations multiply, and information, operational and communications technologies converge. Closely linked to challenges with visibility and alert prioritization, the number three challenge, cited by 49% of respondents, is speed of detection and response.

► What are the three biggest operational challenges for your security team?

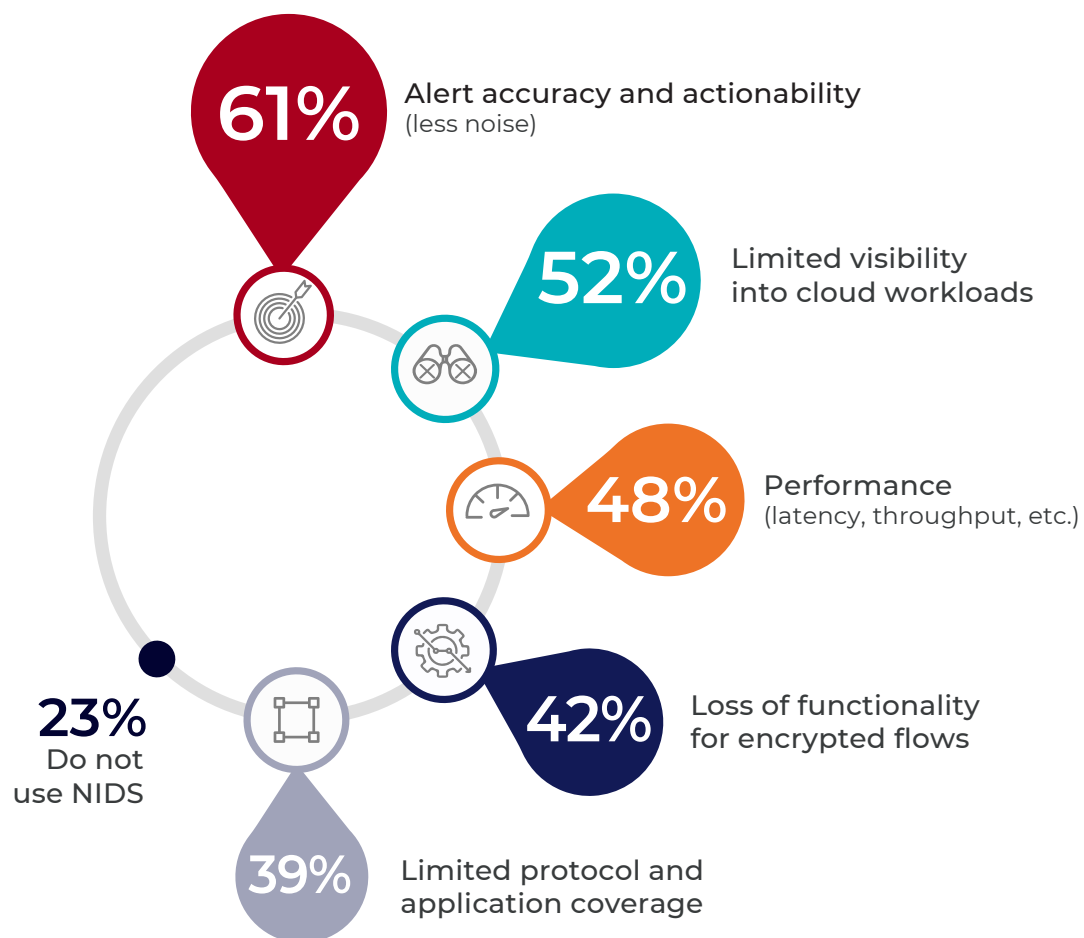


Alert Accuracy & Actionability #1 NIDS/IDS Issue

Echoing the top response for operational challenges, the most pressing need in the specific context of NIDS/IPS deployments is more accurate and actionable alerts (61%). As with effective prioritization of alerts, reducing false positives and alert noise can improve the efficiency and effectiveness of security teams, which would help address the burnout and turnover challenges cited on page 11.

Another difficulty is limited visibility into cloud workloads, cited as the second greatest challenge by 52% of respondents. Technical performance challenges come in at number three (48%), followed by the loss of functionality for encrypted flows (42%) and limited protocol and application coverage (39%). These are all factors respondents cite in explaining why they prefer commercial rather than open source NIDS/IPS solutions (see page 18).

► If you use a NIDS/IPS, what are your greatest challenges?



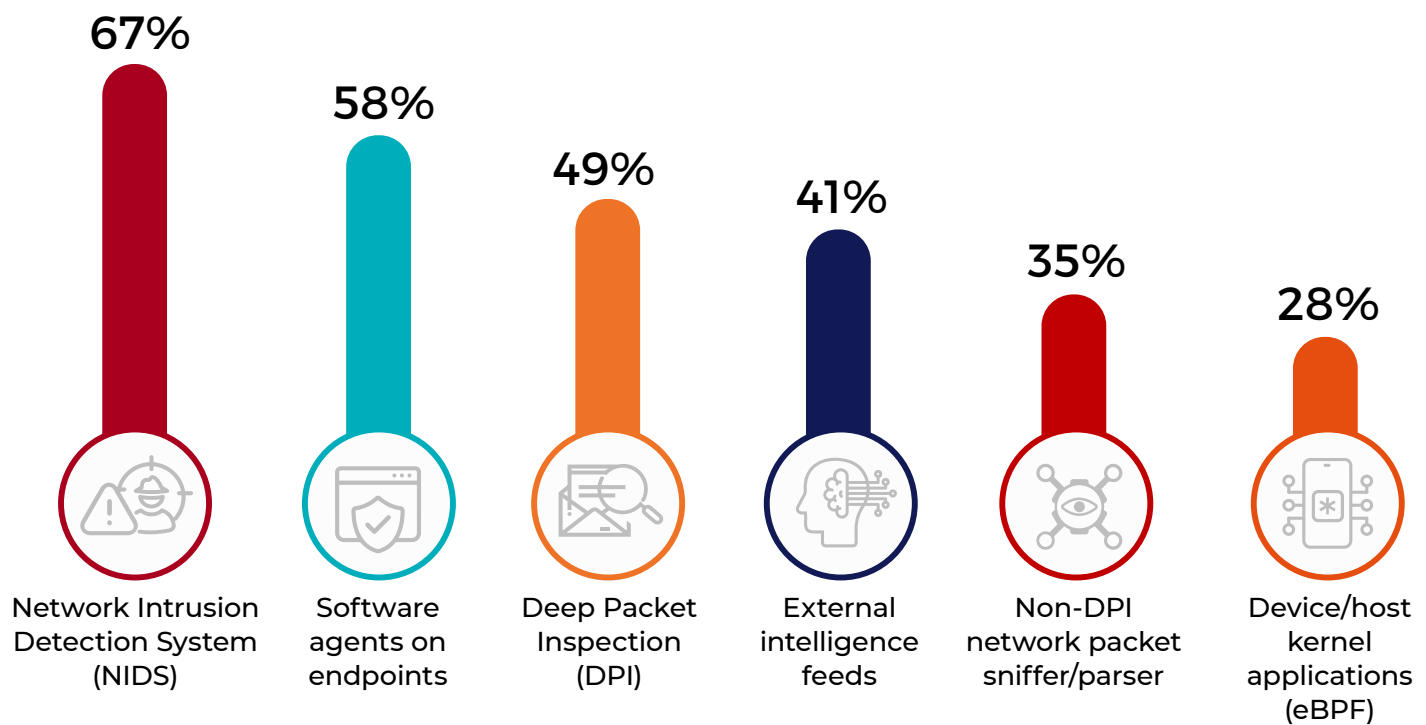
Visibility Challenges Drive Wider Sourcing for Traffic-Related Insights

To address visibility gaps arising from evolving networks, security professionals are turning to an expanded pool of resources for gathering network traffic-related insights. Logically enough, a Network Intrusion Detection System (NIDS) is reported to be the most commonly used tool (67%). Deep Packet Inspection (DPI) (49%) and non-DPI packet sniffers (35%) also make a strong showing, which is to be expected given their long-time leading role in extracting traffic insights.

What is new is relying on sources such as endpoint agents (58%), external intelligence feeds (41%), and device/host kernel applications (eBPF) (28%) to gather network traffic insights (with the latter especially common in cloud workloads).

This reliance on non-network tools for network insights is a two-way street. For example, today advanced DPI can deliver unique insights into devices and users in addition to network flows. This diversification of resources used for cross-domain insights is a welcome development as important strategies such as zero trust and defense-in-depth rely heavily on broadly sourced contextual data to be effective.

► Which types of tools (in embedded or standalone form) do you use to gather network traffic-related insights?

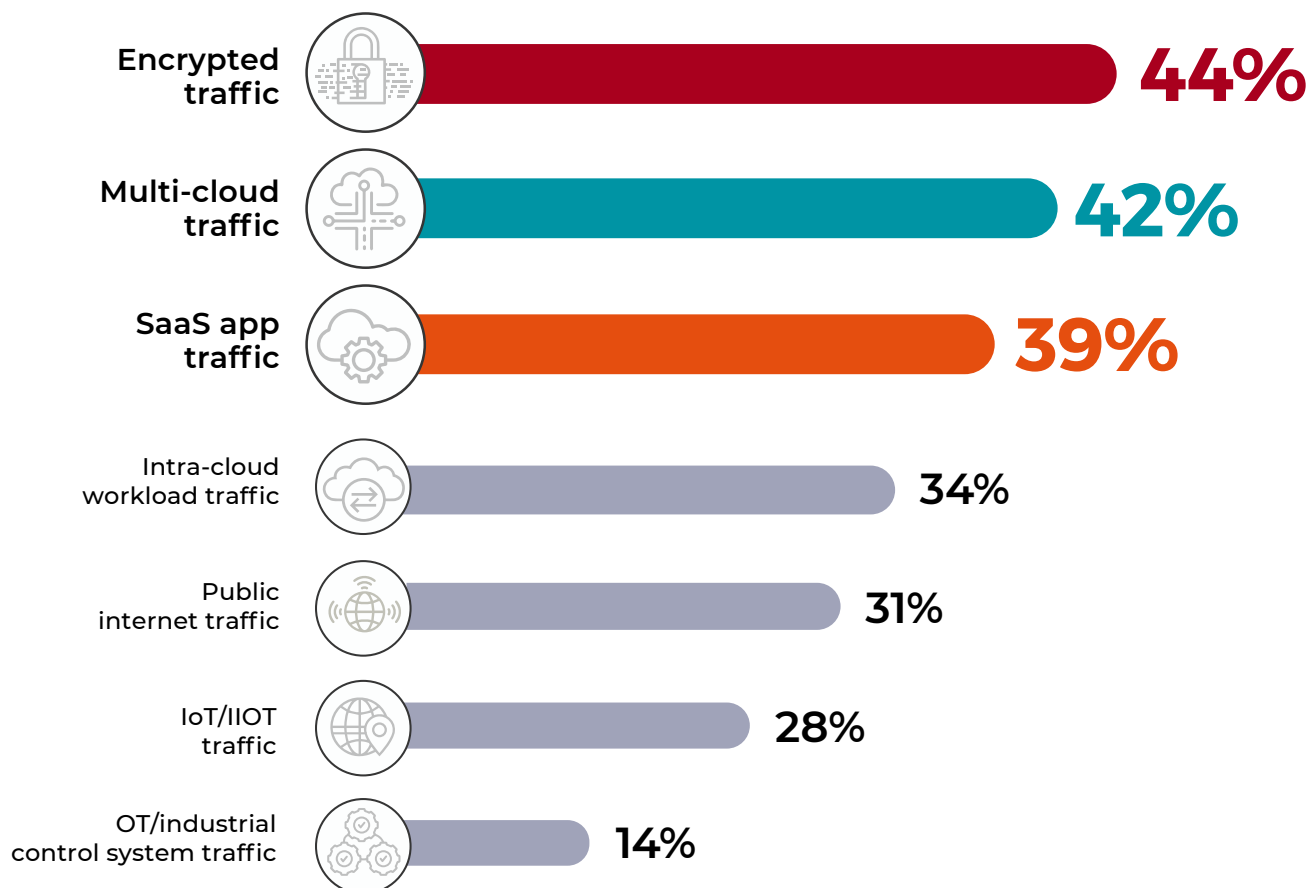


Encrypted Traffic Is the Most Significant Blind Spot

Among specific visibility gaps, respondents rank encrypted traffic as number one (44%), followed closely by multi-cloud traffic (42%) and SaaS app traffic (39%). Cloud and SaaS app use poses a double challenge to visibility: the growth rate outpaces the ability to integrate the apps into monitoring tools and structural challenges make it difficult to extract insights from resources controlled by third parties. Ranked fourth is intra-cloud workload traffic (34%), which underscores the fact that this internal traffic often falls outside the purview of traditional security tools.

Additional sources of concern are public internet traffic (31%) (a challenge due partly to the increase in remote work), IoT and IIoT traffic (28%), and OT/industrial control system traffic (14%), where specialized devices and protocols make visibility and threat detection more difficult. These environments are also often more sensitive to disruptions, making it harder to inspect traffic without impacting operational performance.

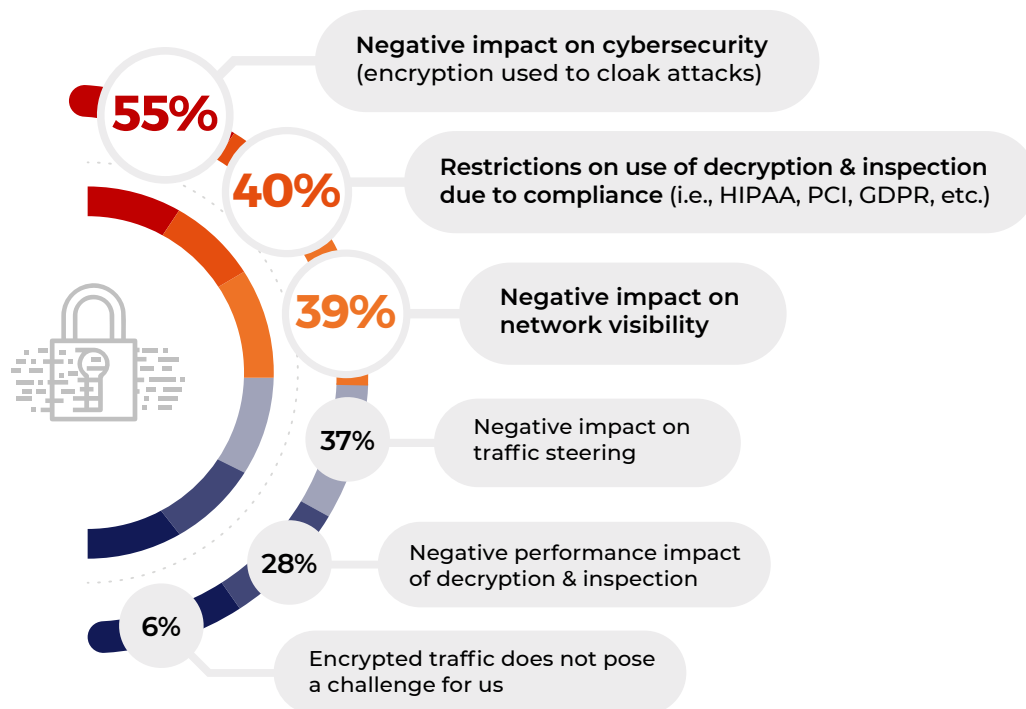
► Where do you feel you have the most significant network traffic visibility gaps?



Encryption Has a Negative Impact on Security

Beyond the negative impact on visibility, encrypted traffic creates many challenges for security (and networking) teams. Ironically, though encryption was developed to strengthen security, respondents report that their number one challenge with its use is the negative impact it has on cybersecurity (55%). Trying to navigate the regulatory issues that govern encryption is the second most significant challenge for respondents (40%), while a close 39% circle back to the recurring theme of visibility impediments, with 37% also reporting that encryption has a negative impact on traffic steering. Additionally, 28% of respondents highlight performance degradation caused by decryption and inspection processes. This highlights a challenge with what could otherwise be a solution to visibility difficulties: decrypting and inspecting all traffic (within the limits of regulations). This strategy is commonly employed by SASE and SSE vendors, who recreate high-performing central gateways on cloud perimeters.

► What challenges do you have with encrypted traffic?



In any case, 11% report the formidable challenge of performing network threat detection on encrypted traffic alone, and 57% perform it on both encrypted and clear traffic.

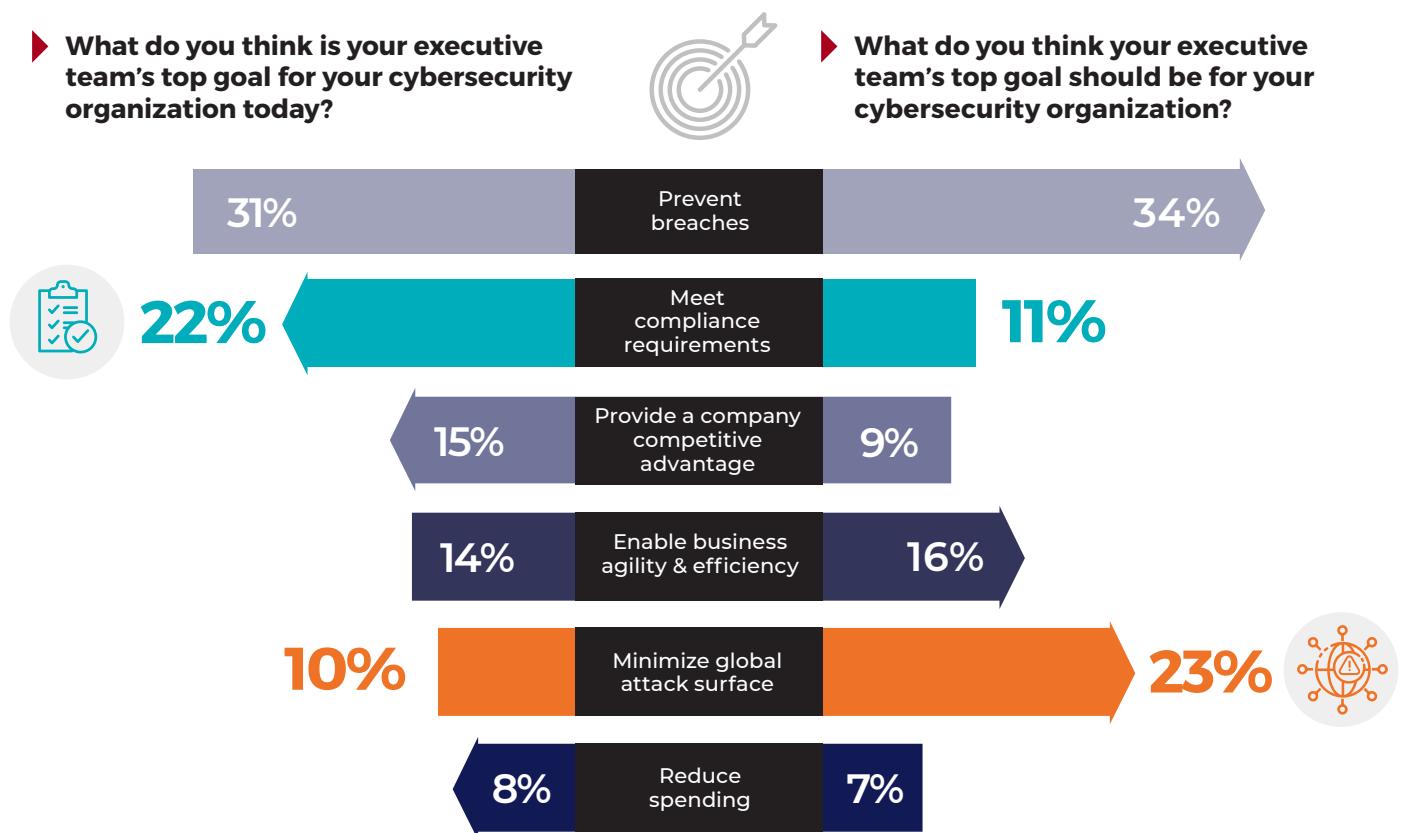
► Do you perform network threat detection on clear or encrypted traffic?



Reducing Attack Surface Should Be Higher Priority

Another indicator of the importance security teams place on closing visibility gaps is the divergence between what security teams think executive priorities are for the security organization versus what security teams think they should be.

Here, security professionals think executives consider meeting compliance requirements as the security organization's number two priority. However, they believe minimizing the global attack surface should actually occupy that spot (with minimizing the global attack surface being dependent on network visibility).



Security Teams Feel Unprepared & Overwhelmed

The top organizational challenge cited by respondents is inadequate in-house skills and training, followed closely by staff burnout and turnover.

Given the high importance respondents placed on AI integration in network threat detection solutions (see page 13), it is likely staff have confidence that one of AI's benefits will be to make them feel better equipped to meet ever more sophisticated attacks. And successfully addressing the top operational challenges of alert fatigue and poor attack surface visibility – also likely with AI support – could certainly be expected to reduce staff burnout and turnover.

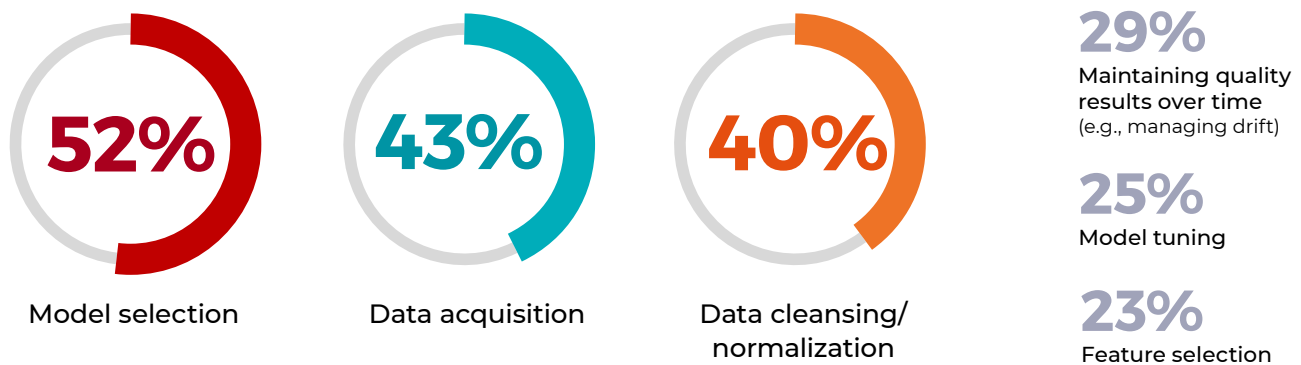
► What are the three biggest organizational challenges facing your security team?



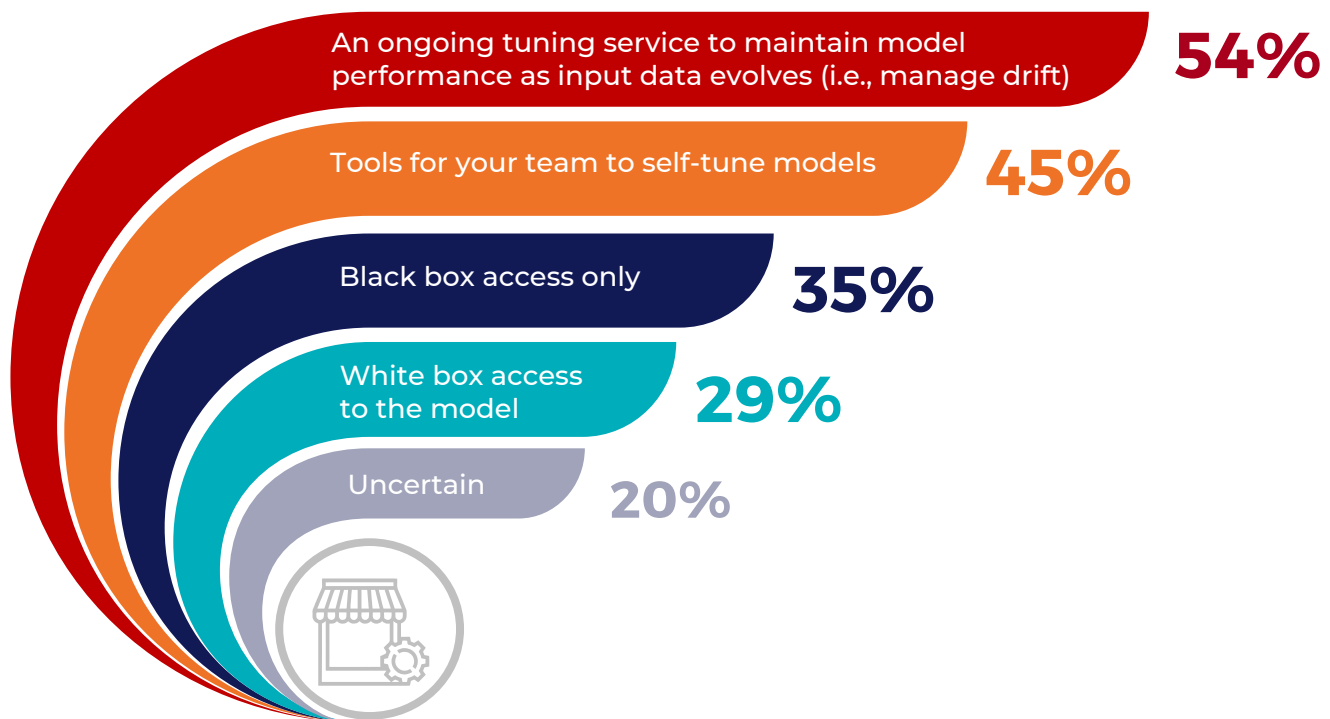
Challenges with ML/AI-Based Network Threat Detection

Of those who use ML/AI, the number one challenge cited is model selection, followed by data acquisition and data cleansing and normalization. Regarding the 4th and 5th challenges, managing drift and model tuning, vendors are providing more tools to empower users to address these natural AI lifecycle evolutions on their own, though more than one third (35%) still provide only black box access to their ML/AI solutions.

► For ML/AI-based network threat detection, which of the following do you find challenging?



► If you currently have a commercial network threat detection product that uses AI/ML for advanced threat detection, what access and support does the vendor provide?



Very High Confidence in AI's Value

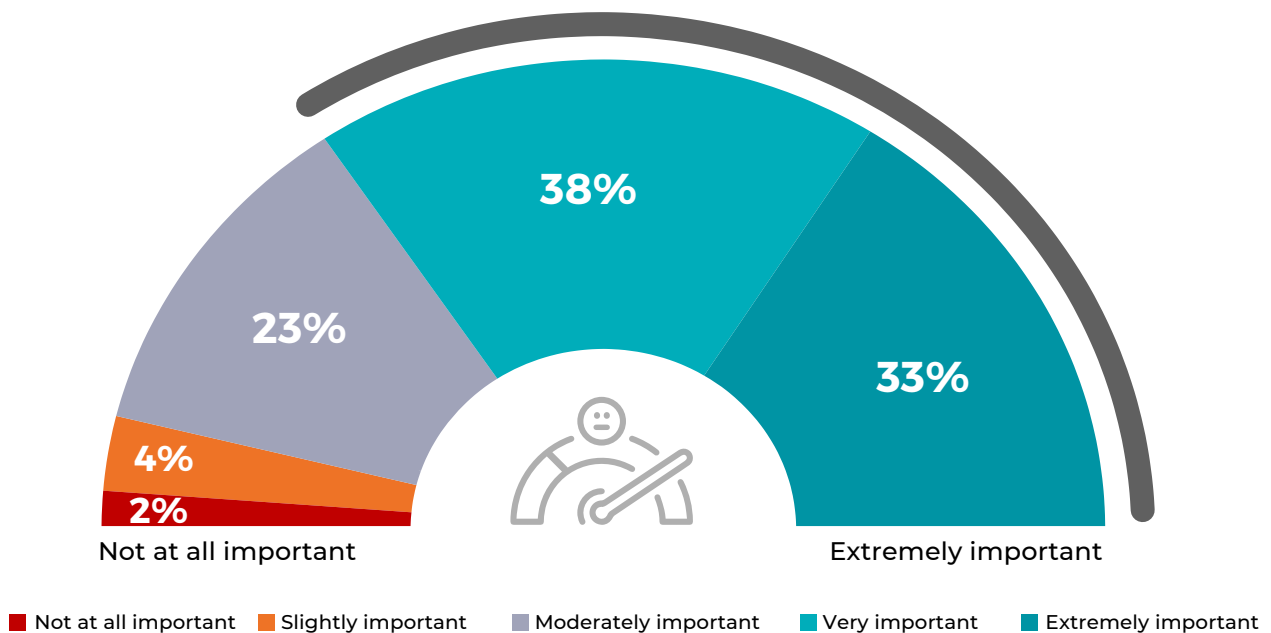
A striking 71% of respondents consider it very (38%) or extremely (33%) important for network threat detection to incorporate AI. Another 23% consider it moderately important, with only 6% considering it slightly important (4%) or not important (2%).

Part of this confidence may be tied to AI's ability to rapidly analyze large volumes of network traffic and detect subtle patterns or anomalies—especially within encrypted or highly complex traffic—that are indicative of sophisticated attacks (which, in turn, increasingly employ AI).

However, given that the three top operational challenges for security teams are 1) the difficulty of prioritizing alerts, 2) no (or poor) visibility into the global attack surface, and 3) unsatisfactory speed of detection and response, it is logical to assume that security teams have faith that AI can be used to address a wide variety of challenges.

► How important do you think it is for network threat detection to incorporate AI for advanced threat detection?

71% of respondents think it is very or extremely important to incorporate AI for advanced threat detection

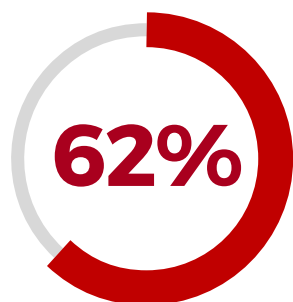


Automatic Threat Scoring & Prioritization Most-Valued Capability

Respondents place automation and simplification at the top of their must-have capabilities for network threat detection solutions. 62% of respondents see automatic threat scoring and prioritization as a must-have, while 59% value correlation of relevant data, events, and alerts into single incidents. Close behind, 57% desire automated and/or guided response processes, and 53% want their solution to automatically add contextual data to alerts.

Against this backdrop of a deep desire for automation, it is interesting to note that generative-AI (or GenAI) assistance, which involves a collaborative dialogue between the security analyst and the AI application, comes near the end of the must-haves. It is an indicator, perhaps, that full automation is now valued more highly than interactive assistance.

► What capabilities do you think are must-haves for a good network threat detection solution?



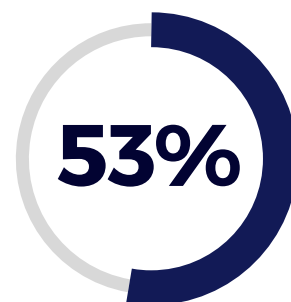
Automatically score
and prioritize threats



Correlate relevant
data/events/alerts
to single incidents



Automate and/or
guide response
processes



Automatically
add contextual
data to alerts

48%

Integrate external
threat intelligence
sources

41%

Support threat
hunting and
forensics

35%

Block known
threats

31%

Generate
MITRE ATT&CK
mappings

28%

Incorporate
generative-AI
assistance

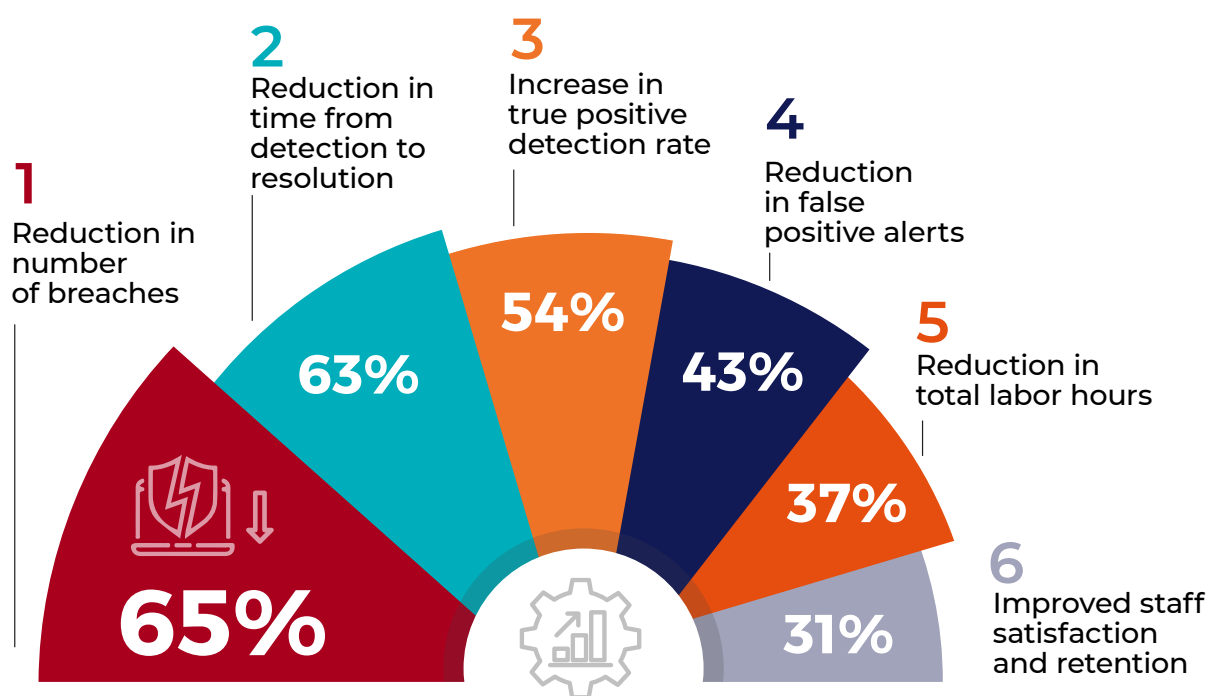
14%

Support offline
packet capture
(pcap) processing

Reduction in Breaches Tops KPI List

Respondents consider the reduction in the number of breaches as the most useful KPI for judging threat detection effectiveness. In a network threat detection context, this does not mean blocking threats at the perimeter, but rather finding and stopping infiltrations before data is accessed and released, exfiltrated, or encrypted. And for this, one has to be aware of breaches in order to measure their reduction over time, hence high rankings of reducing time from detection to resolution (63%), increasing true positive detections - i.e., not missing actual threats (54%), and reducing false positives (43%), which take valuable time away from finding and stopping legitimate threats.

► **What metric(s) do you think are most useful for evaluating the efficacy of a network threat detection solution?**



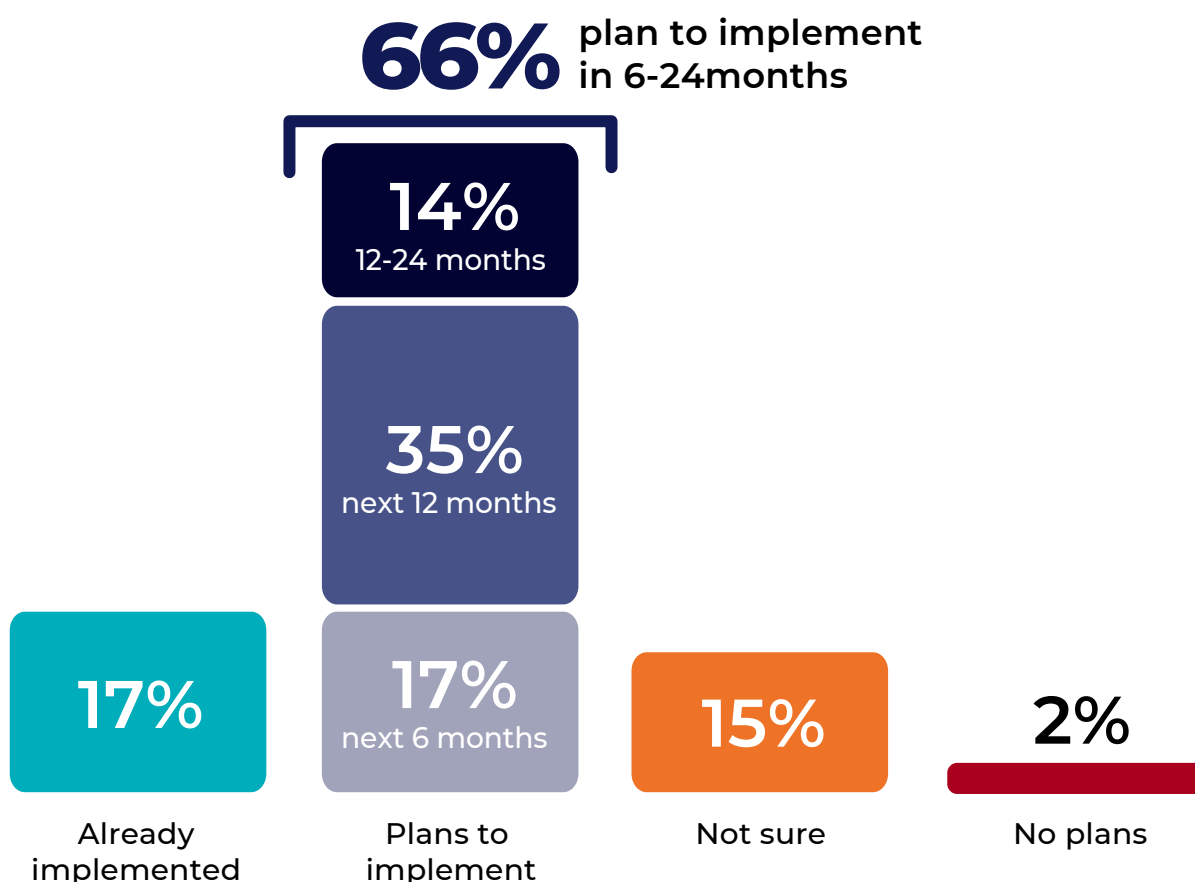
Broad Expansion for Anomaly Detection

Network intrusion detection systems use two principal techniques for identifying breaches. One analyzes traffic for specific patterns, or signatures, of known threats, while the other looks for anomalous behaviors. The latter typically works by creating a baseline of what normal (safe) traffic looks like, and then uses statistical and/or machine learning to detect anomalies indicative of a breach or vulnerability.

Anomaly detection is used to a limited extent in conventional IDS/IPS but is a key pillar of NDR solutions. It offers a more effective method of catching advanced threats than signatures, as hackers rapidly adapt their techniques once an attack method is exposed and codified via a signature.

Reflecting confidence in this capacity to catch advanced attacks, 83% of all organizations say they either currently use anomaly detection (17%) or plan to do so over the next 6-24 months (66%). 15% are uncertain of their organization's intent to use it. Only 2% report no plans for using anomaly-based network threat detection.

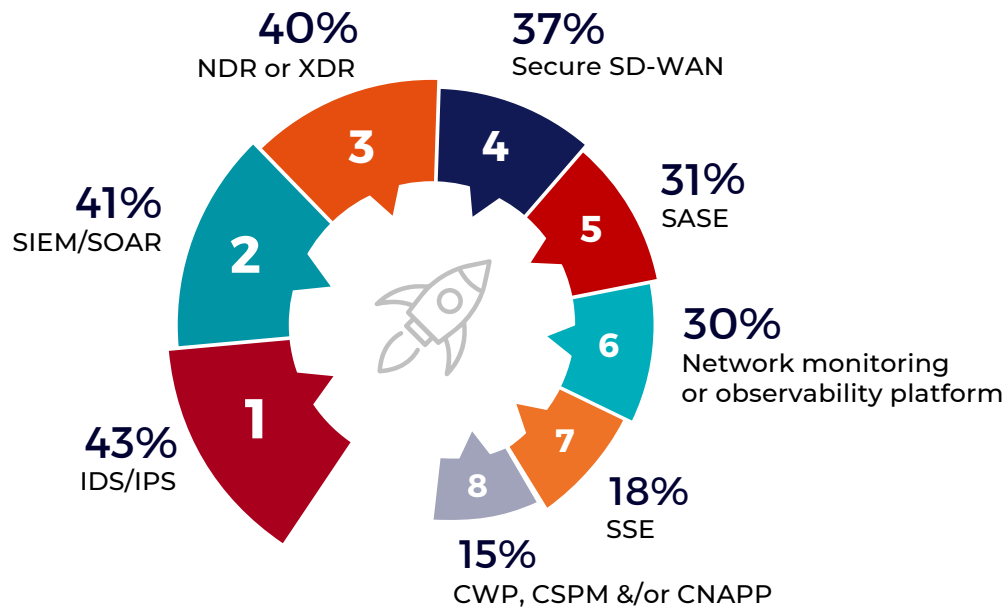
► What are your plans regarding anomaly-based network threat detection capabilities in your organization?



IDS/IPS & Specialized NTD Tools Are Popular Choices

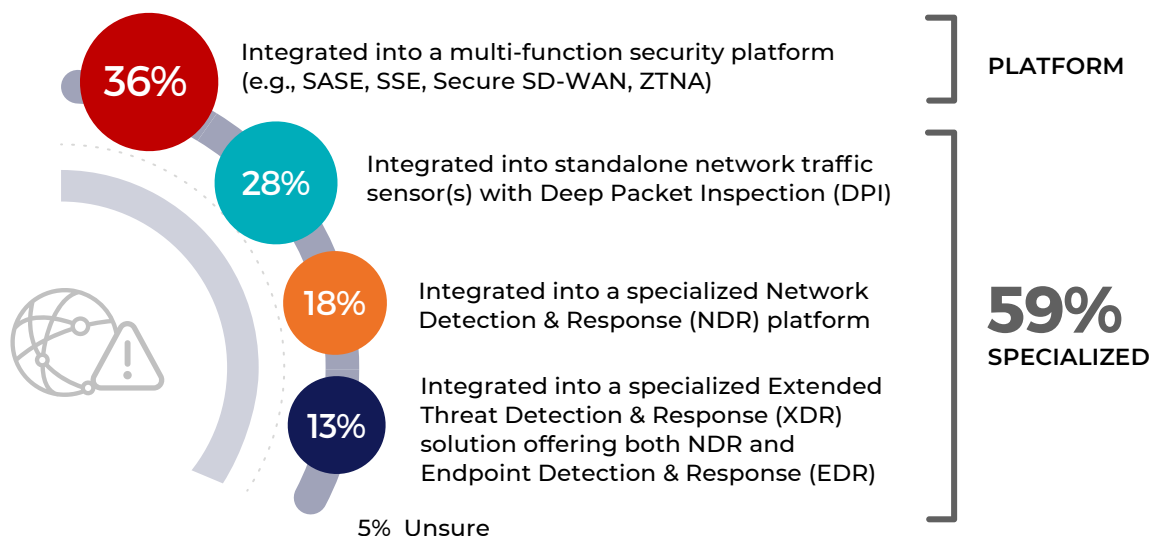
IDS/IPS is currently the most widely deployed network threat detection tool (43%). Two other specialized threat detection tools, SIEM/SOAR and NDR/XDR, are more widely deployed than broader platforms like Secure SD-WAN, SASE and SSE.

► Which of the following are currently deployed at your organization?



Furthermore, per the second question below, only a minority (36%) consider integration into a broader, multi-functional security platform to be the most effective option for their organization, while 59% cite one of three types of specialized NTD solutions (DPI-based NTA sensor, NDR, or XDR). This may change as SASE and SSE adoption continues to grow, but it would not be surprising to see continued deployment of best-of-breed NTD solutions alongside such platforms.

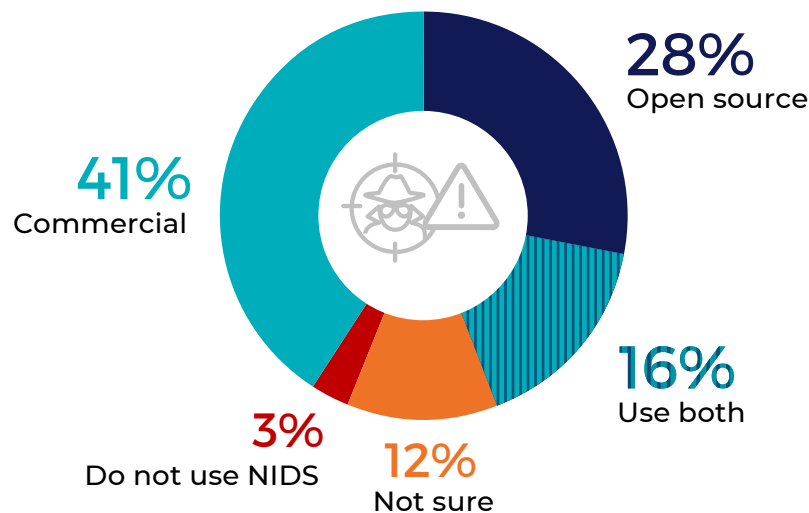
► Which type of network threat detection deployment would be most effective for your organization?



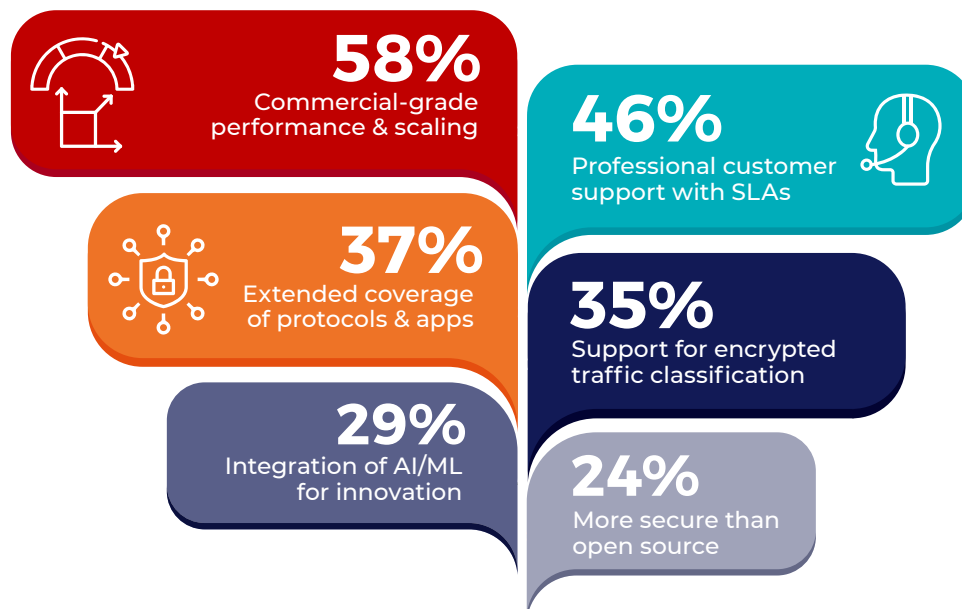
Commercial NIDS Preferred over Open Source

Security professionals express a preference for commercial over open source solutions (41% vs 28%), though 16% use both. The top three reasons for the commercial preference are performance and scalability, customer support, and protocol coverage. It is important to note, however, that most commercial NIDS/IPS are built upon an open source NIDS/IPS foundation. For example, the Enea Qosmos Threat Detection SDK was developed in partnership with the Open Information Security Foundation (OISF, Suricata's maker). It tightly integrates core functionalities from Suricata with Enea's deep packet inspection engine, the Enea Qosmos ixEngine®, to help solution developers meet the unique performance demands of commercial-grade deployments.

► If you use a Network Intrusion Detection System (NIDS), is it open source or commercial?



► What do you see as the top three reasons to use a commercial NIDS solution?

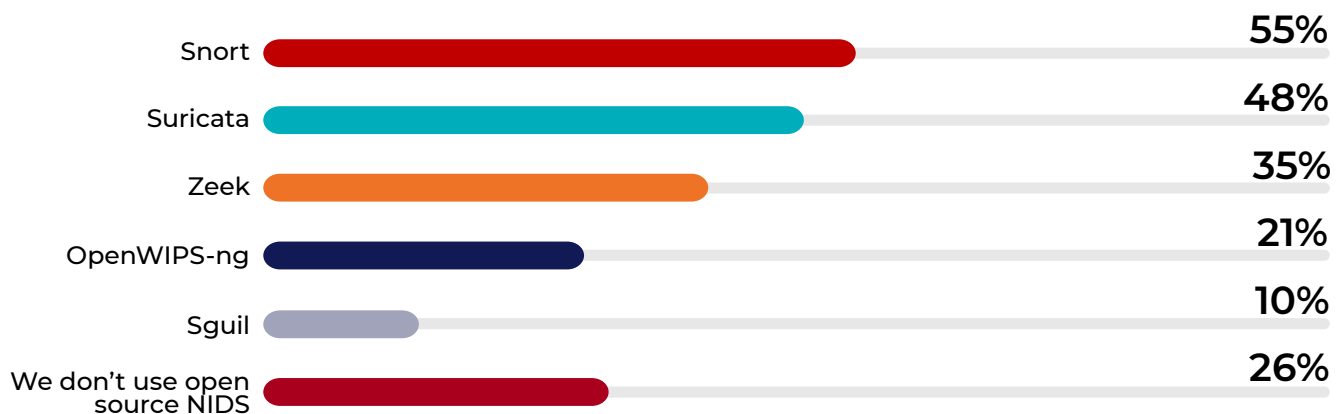


Snort & Suricata Most Popular Open Source NTD Tools

Snort is cited as the most frequently used open source NIDS, followed closely by Suricata. The number three most commonly cited NIDS is Zeek. These tools have been around for a long time, and all continue to evolve and to play an important role in protecting networks worldwide.

Created in 1998, Snort was originally developed as a packet sniffer and logger and evolved to support signature- and anomaly-based intrusion detection. First released in 2010, Suricata was originally developed as a signature-based NIDS/IPS, but over time has added some anomaly detection and network security monitoring capabilities. First deployed in 1995, Zeek is a network security monitoring tool but can be used to provide some NIDS functionality.

► If you use an open source NIDS, which one?

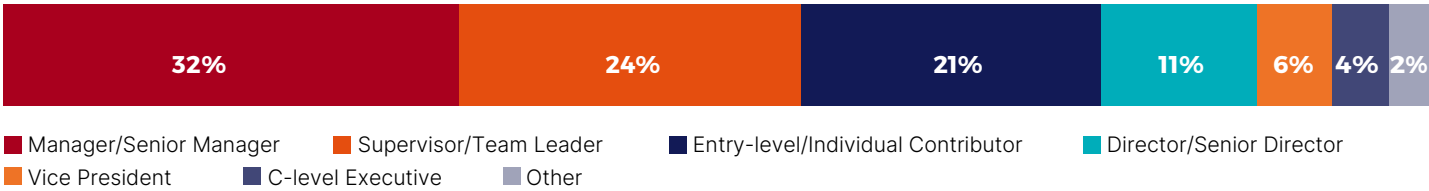


Methodology and Demographics

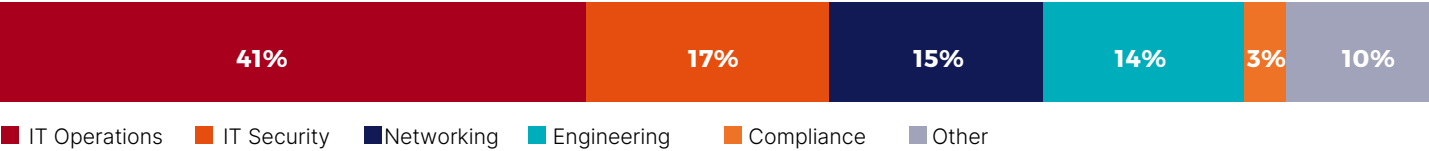
This 2024 Network Threat Detection Report is based on a comprehensive online survey of 327 cybersecurity professionals, conducted in September 2024, to gain deep insight into the latest trends, key challenges, and solutions for network threat detection.

The survey utilized a methodology ensuring a diverse representation of respondents, from technical executives to IT security practitioners, across various industries and organization sizes. This approach ensures a holistic and balanced view of the insider threat landscape, capturing insights from different organizational perspectives and experiences.

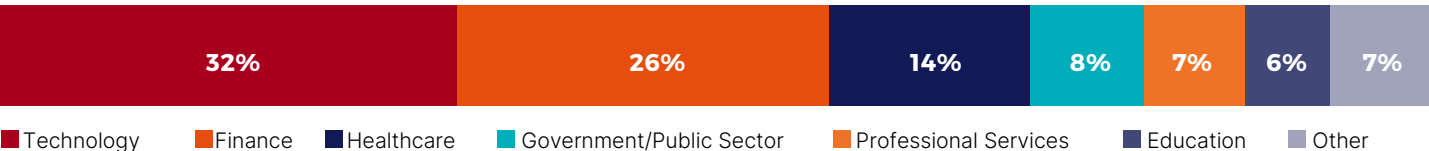
CAREER LEVEL



DEPARTMENT



INDUSTRY



COMPANY SIZE



Reuse of content

We encourage the reuse of data, charts, and text published in this report under the terms of this [Creative Commons Attribution 4.0 International License](#). You're free to share and make commercial use of this work as long as you attribute the report as stipulated in the terms of the license. For example: "2024 Network Threat Detection Report by Cybersecurity Insiders and Enea."



About Arista Networks

Arista Networks is an industry leader in zero trust networking, delivering security and observability across wired, wireless, and cloud infrastructure. Arista AVA™, an AI decision support system, enables an integrated suite of security platforms for standards-based network access control, autonomous threat hunting, and identity-aware microsegmentation. Importantly, these zero trust platforms are built on network infrastructure powered by Arista EOS™ and NetDL™, avoiding network security overlays and thus reducing costs while accelerating zero trust maturity and lowering breach impact.

Arista Networks has been recognized as a market leader by Gartner, Forrester, and KuppingerCole, among others.

arista.com/security



About Enea

We are a world-leading specialist in advanced telecom and cybersecurity software with a vision to make the world's communications safer and more efficient. As the most widely deployed Deep Packet Inspection (DPI) technology in cybersecurity and networking solutions, the Enea Qosmos products classify traffic in real-time and provide granular information about network activities. Enea also offers IDS-based threat detection capabilities as an SDK, enabling easy and tight integration with cybersecurity solutions while remaining highly flexible and scalable. Enea is headquartered in Stockholm, Sweden and is listed on NASDAQ Stockholm.

enea.com/dpi-tech



About Custocy

Custocy is a French spin-off from IMS Networks, specialised in cybersecurity software. Based in Toulouse, in the Occitanie region, it has a Research and Development team of around fifteen PhDs and engineers who have been developing an artificial intelligence engine since 2019. This engine is integrated into a SaaS platform for Network Detection and Response. Custocy has established a high-level collaboration with the LAAS-CNRS laboratory. Custocy is a laureate of the i-NOV innovation competition as part of the French government's France 2030 plan and Bpifrance. In May 2024, Custocy was named "Product of the Year" at the Paris Cyber Show.

custocy.ai

Cybersecurity

I N S I D E R S

Cybersecurity Insiders brings together 600,000+ IT security professionals and world-class technology vendors to facilitate smart problem-solving and collaboration in tackling today's most critical cybersecurity challenges.

Our approach focuses on creating and curating unique content that educates and informs cybersecurity professionals about the latest cybersecurity trends, solutions, and best practices. From comprehensive research studies and unbiased product reviews to practical e-guides, engaging webinars, and educational articles - we are committed to providing resources that provide evidence-based answers to today's complex cybersecurity challenges.

For more information:

email us info@cybersecurity-insiders.com or visit cybersecurity-insiders.com